IoT (Internet of Things)

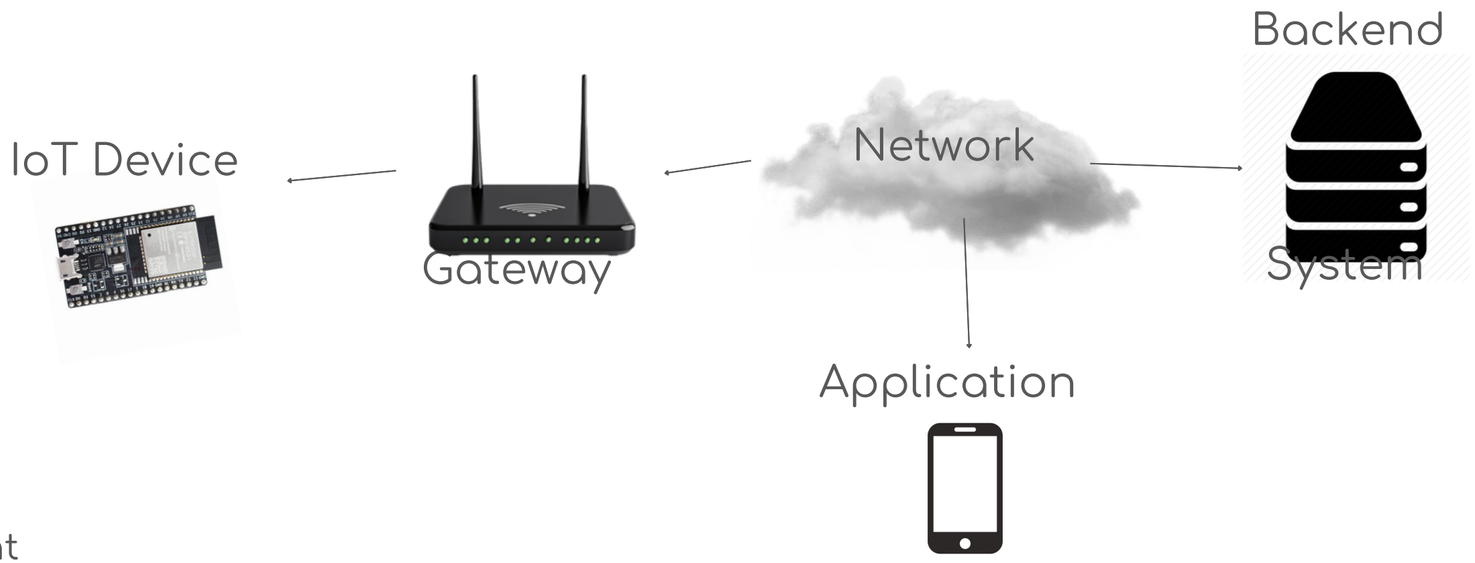
Environment Data Monitoring

What is IoT?

(Internet of Things)

connect every physical thing —using internet→ decide better what to do

1. Device hardware
   - SoC
   - Sensor
   - MCU
   - Actuator
2. Device firmware
   - control hardware
   - stored on ROM/flash memory
   - run on SoC
3. Communication
   - sharing data
   - protocol
4. Backend System
   - where all data is collected
   - Management
   - Monitoring
   - Big data analytic
5. End User application
   - Interface for user
   - Web
   - Mobile app
   - Voice assistant



Why IoT?

Better decision Making

Maximizing efficiency

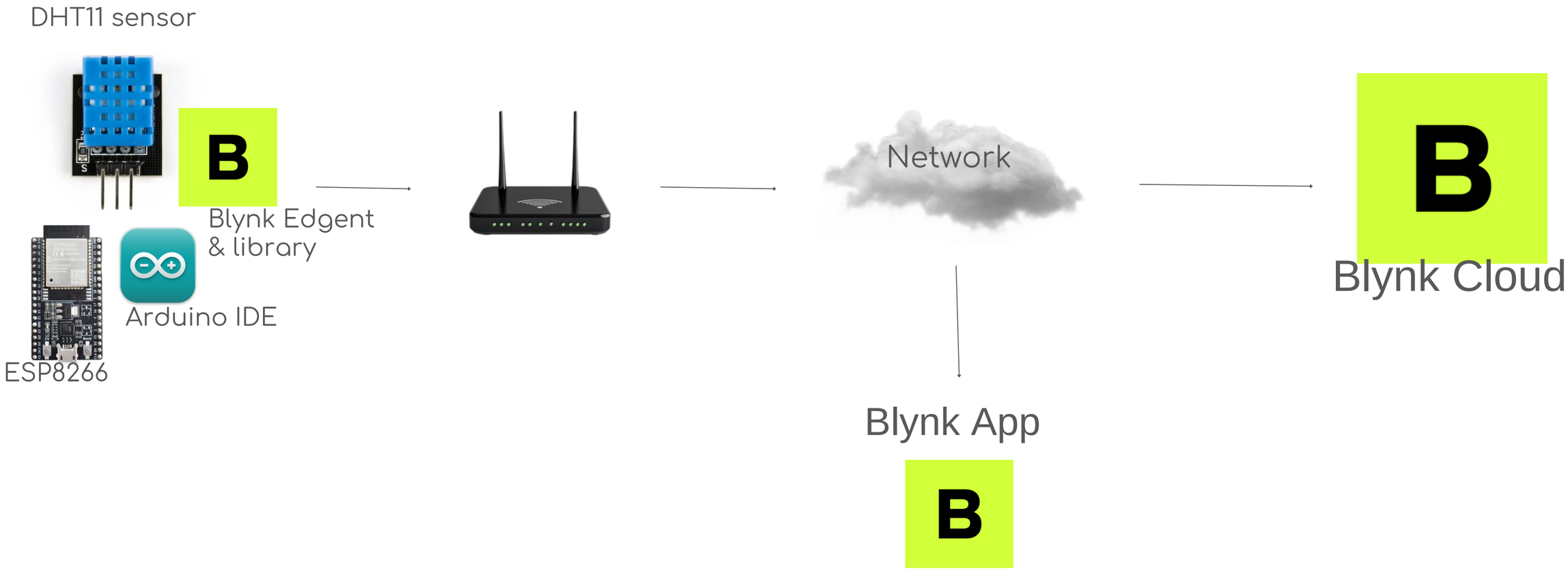
Reducing Cost

Safety & risk reduction

to improve the quality of human lives

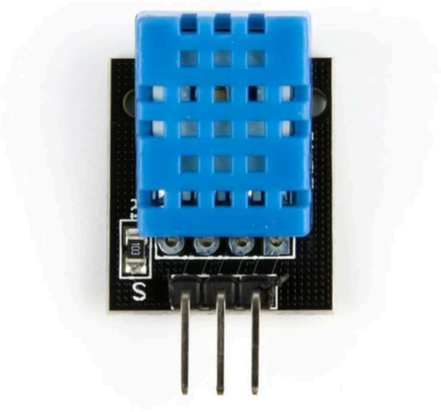
Environment Data Monitoring

Humidity & Temperature



Device Hardware

DHT 11



Measure temperature
& humidity

0°C - 50°C

20% - 80%

1Hz

ESP 32

SoC

Only chip

Modules

Chip works essentials
(small pcb + antenna)

Board

Microcontroller
(module + voltage regulator + button)

Type of ESP

ESP 8266
legacy

C Family
General purpose

H Family
Smart Home

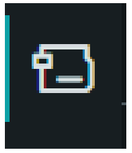
S Family
Machine learning & AI

Development tools

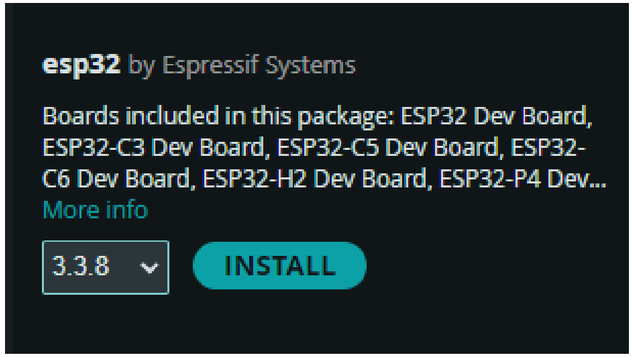
Arduino IDE

Flashing blynk firmware to ESP

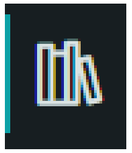
1. Setup board setting

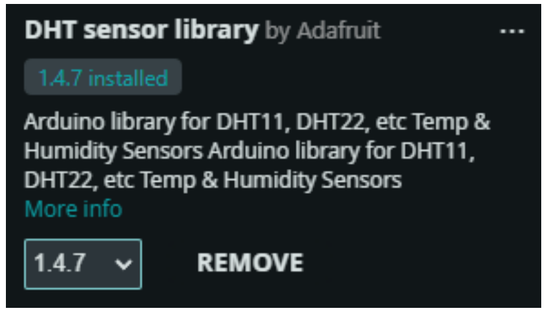


Additional boards manager URLs: http://arduino.esp8266.com/stable/package_esp8266com_index.json



2. Install libraries





3. Verify and UPload

IoT solution

Blynk

Bynk.Console
Web Application

Bynk.Apps
Mobile Application

Bynk.Edgent & library
Device provisioning

Getting started

1. Sign up
2. Choose blueprint & flash firmware
3. Activate device
4. Create dashboard

Blynk Firmware Structure

File	What it does?
.ino	Main program: Start point of project
BlynkEdgent.h	Boot, connect, run modes
BlynkState.h	running, etc
ConfigMode.h	WiFi provisioning
ConfigStore.h	Memory saver: wifi & login details
Console.h	Debug console
Indicator.h	Status feedback
OTA.h	Firmware updates
ResetButton.h	Hardware reset logic
Setting.h	Setting files: name, options

Program structure

main project .ino files

1. Define identification variable

```
#define BLYNK_TEMPLATE_ID "xxxxxxxxxxxx"
#define BLYNK_TEMPLATE_NAME "some random name"
```

2. Declare necessary library

```
#include <DHT.h>
#include "BlynkEdgent.h"
```

3. Define peripheral variable

```
#define DHTPIN 26
#define DHTTYPE DHT11
```

```
void loop() {
  BlynkEdgent.run();
  timer.run();
}
```

4. Define Function

```
void readAndSendDhtData() {
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();
  if (isnan(humidity) || isnan(temperature)) {
    Serial.println("Failed to read from DHT");
  } else {
    float humidityDelta = abs(humidity - DHT_HUMIDITY) - DHT_HUMIDITY_IGNORED_DELTA;
    float temperatureDelta = abs(temperature - DHT_TEMPERATURE) - DHT_HUMIDITY_IGNORED_DELTA;
    if (humidityDelta > 0 || temperatureDelta > 0) {
      DHT_HUMIDITY = humidity;
      DHT_TEMPERATURE = temperature;
      Serial.printf("Humidity: %f%%. Temperature: %f*C.\n", humidity, temperature);
      sendDhtData();
    }
  }
}
```

```
void sendDhtData() {
  Serial.println("Sending DHT data");
  Blynk.virtualWrite(DHT_BLYNK_VPIN_TEMPERATURE, DHT_TEMPERATURE);
  Blynk.virtualWrite(DHT_BLYNK_VPIN_HUMIDITY, DHT_HUMIDITY);
}
```

5. Define setup & loop

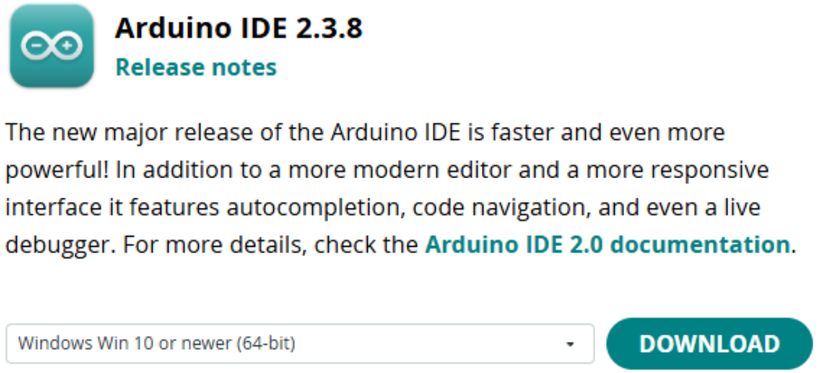
```
void setup()
{
  Serial.begin(115200);
  delay(100);

  BlynkEdgent.begin();

  setupDht();
//  setupBMP();

  // Set up timer to run every 5 sec
  timer.setInterval(5000L, reandAndSendSensorsData);
```

Tutor

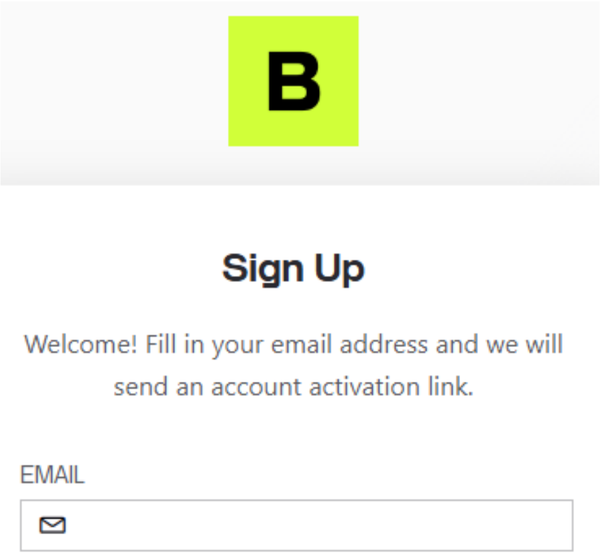
Session

Preparation

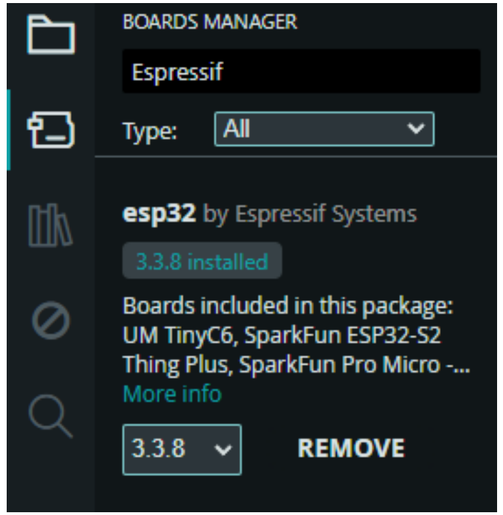
1. Install Arduino IDE



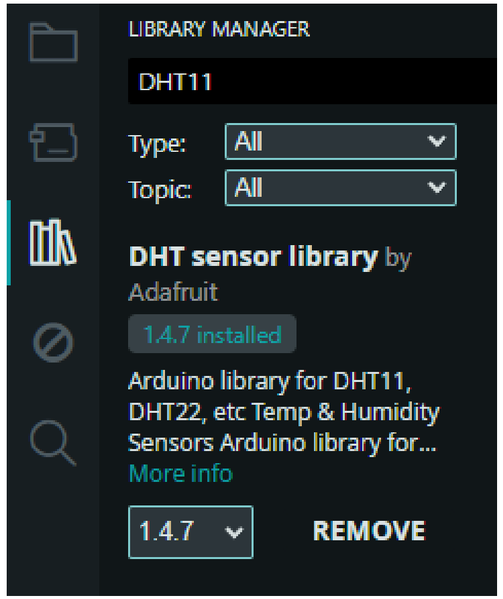
2. Create Blynk account on web



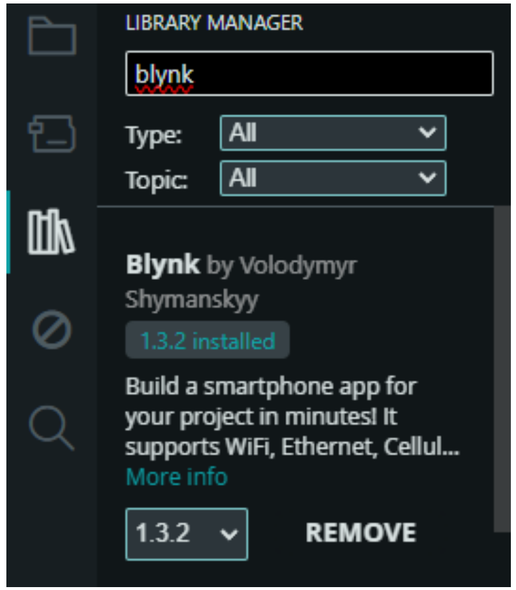
3. Install ESP32 Board Manager



4. Install DHT11 Library

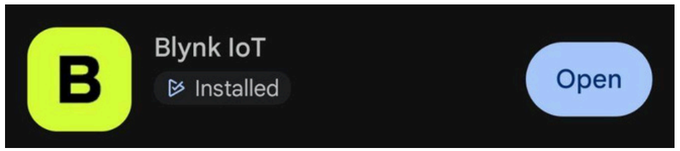


5. Install Blynk Library



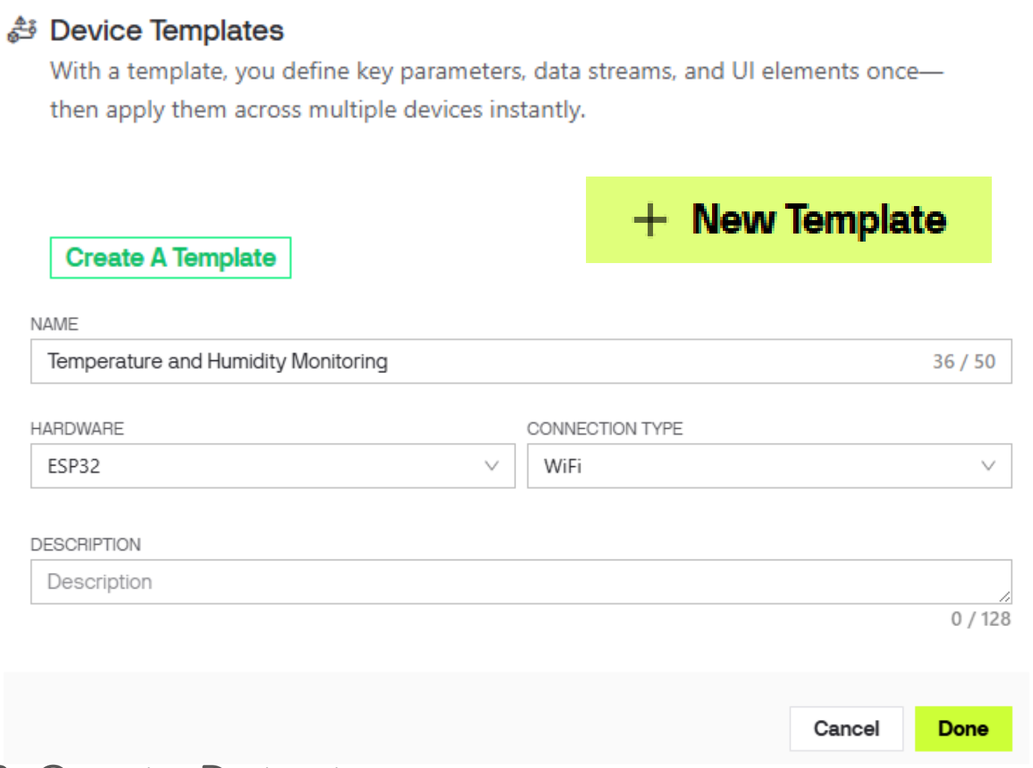
6. Download Firmware

7. Install & login blynk IoT mobile app

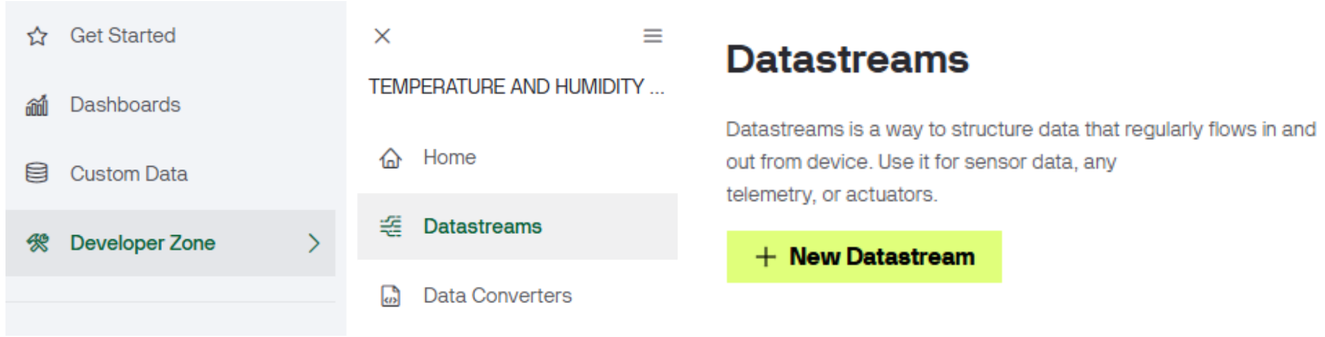


#1 set up blyn k web

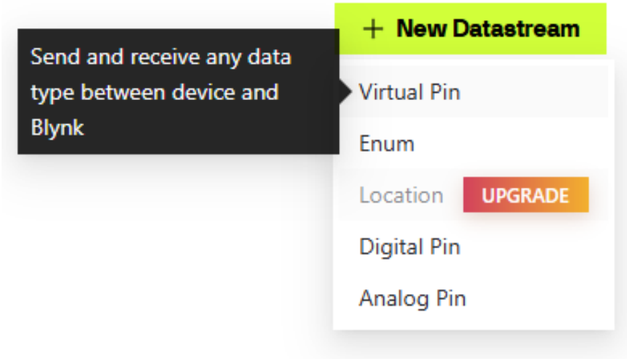
1. Create Template

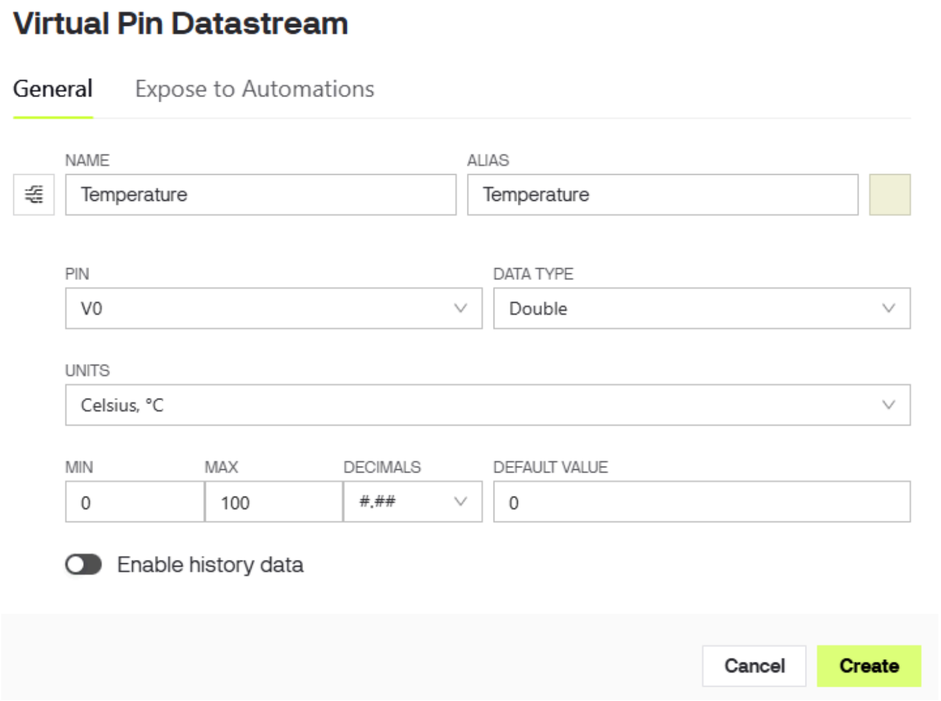


2. Create Datastreams



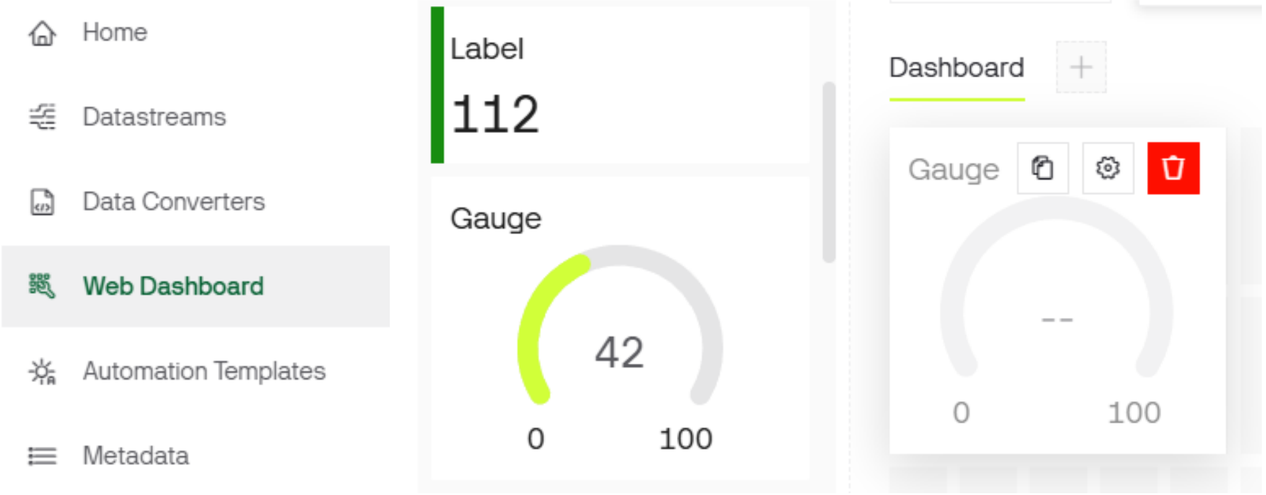
3. Create Virtual pin

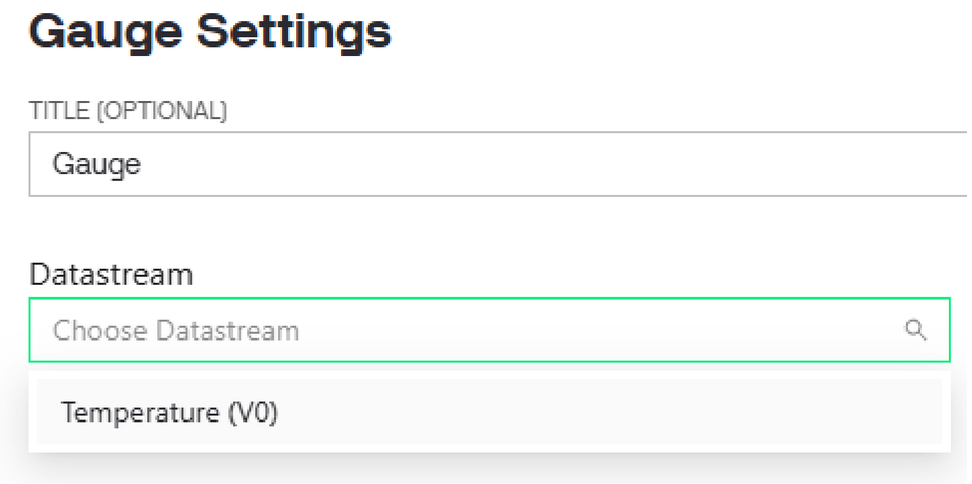




#1 set up blyn k web

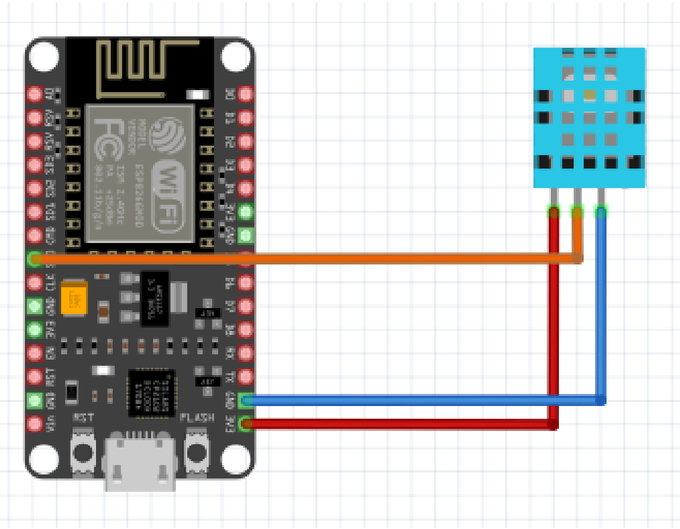
4. Create Dashboard





#2 connect ESP32

1. Wire DHT11 to ESP32



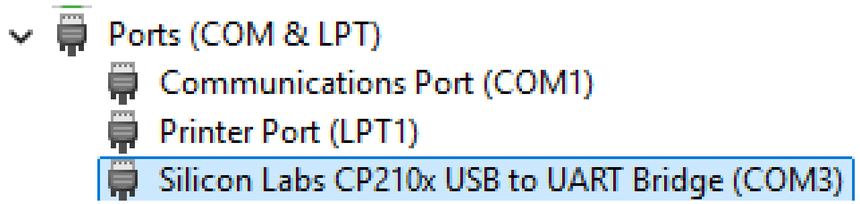
2. Edit template ID on .ino file

Firmware configuration

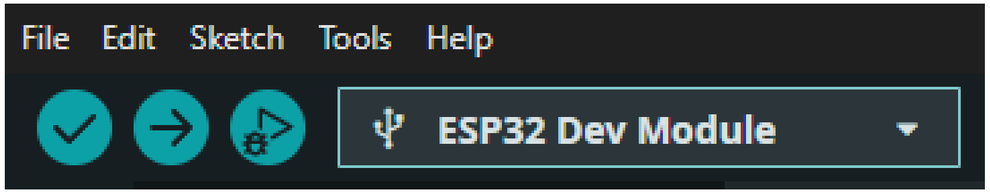
Template ID and Template Name should be declared at the very top of the firmware code.

```
#define BLYNK_TEMPLATE_ID
"TMPL6Cvf_7bkB"
#define BLYNK_TEMPLATE_NAME
"Temperature and Humidity Monitoring"
```

3. Connect ESP32 to PC



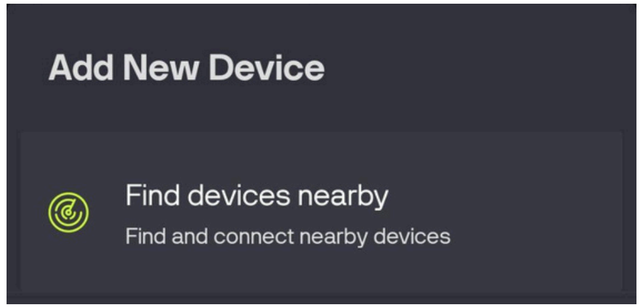
4. Choose board, verify, and flash



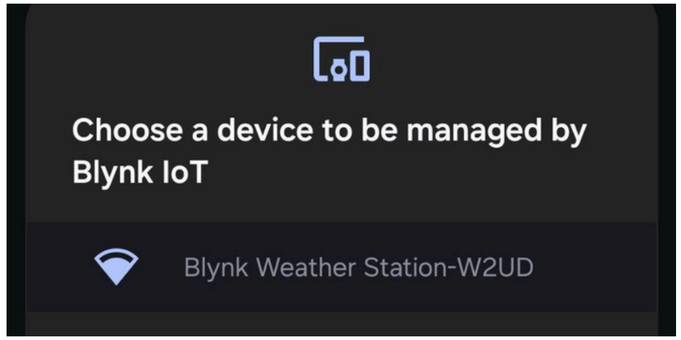

```
#define BLYNK_TEMPLATE_ID "xxxxxxxxxxxxxxx"
#define BLYNK_TEMPLATE_NAME "XXXXXXXXXXXXXXXXXXXX"
```

#3 connect to Blynk IoT Mobile App

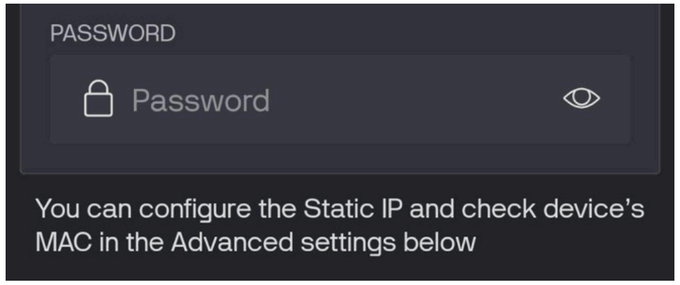
1. Find nearby device



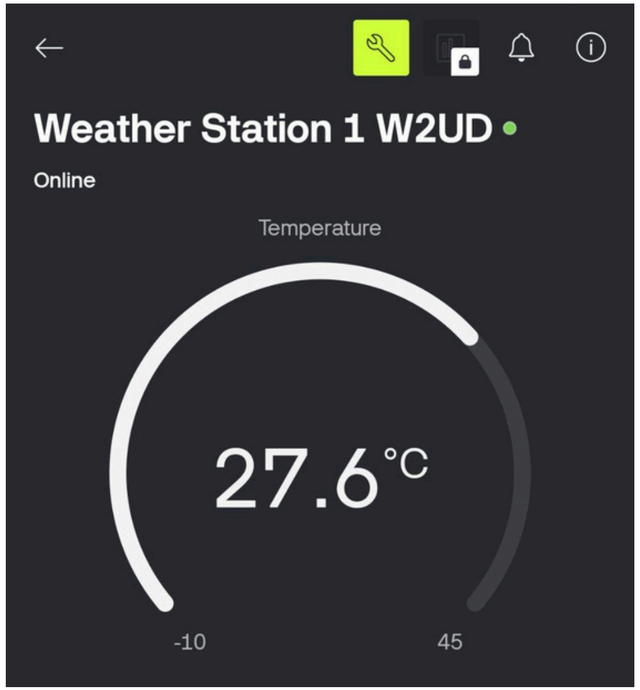
2. Connect device



3. Add Wifi Credentials



4. Blynk IoT Mobile App is connected



FGD

Forum Group Discussion

Case Study

1. Display normal but uncomfortable

Root cause:
limited amount of sensor
& wrong positioning

2. No text on display

Root cause:
data reading mismatch
& no conditional for printing data

Thank You